

Investigating Brexit Implications on the Irish Agri-Food Exports: A Simulation based Scenario Mapping Model

By

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- Tariff Barriers: Applying Tariffs, Quotas, and Rules of Origins on the UK and EU products
- Non-Tariff Barriers Applied at IRE, GB, and the EU ports
 - *Documentary Checks*
 - *Full Identity and SPS Checks*
 - *Security and Immigration Checks*
- Uncertainty of the tradition UK Land-bridge efficiency (Transit Checks)
- Shortage of Checks Infrastructure at the GB Ports

- Political Tension (NI Protocol, fishing rights)
- Collapse in trade volume between the EU states (incl. Ireland) & the UK
- Collapse in freight traffic through Dublin Port, while 43% increase in Rosslare Europort freight traffic
- Increasing direct shipping services to Europe from 3 weekly ferries to 16 weekly ferries
- Importers and hauliers are struggling to cope with the complex new red-tapes in terms of declarations and paperwork
- Lack of necessary customs expertise and knowledge
- Trucks congestions at customs checkpoints & Uncertainty of Land-bridge
- Agri-food Exports haven't experienced harmful effects yet

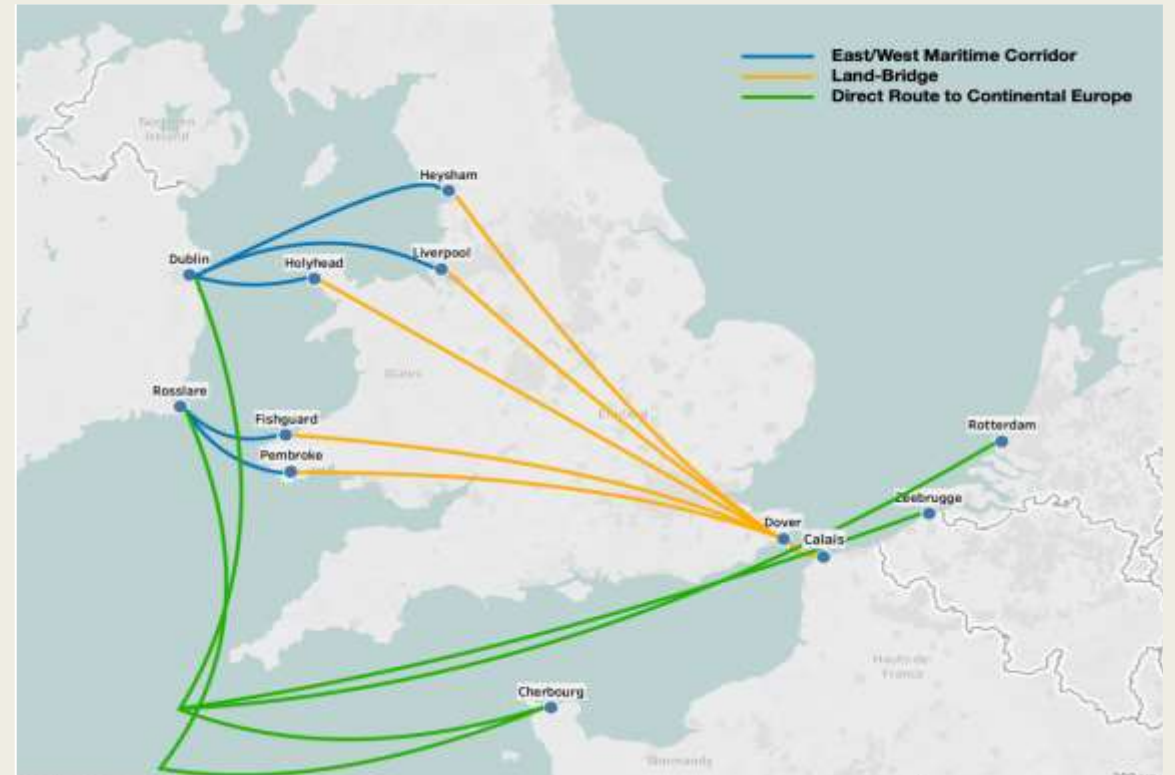
Brexit Disruption & Business Responses



Business Case	Challenges	Actions
Irish Importers	• New Customs Cost • Longer Shipping Time would add to the Cost	• Investigate whether to move manufacturing to Ireland or shift markets • Assess the impact of tariffs and customs charges on total Cost and Profitability • Negotiate with Suppliers to Share Increased Cost
Irish Exporters	• Higher Cost for Importing Raw Materials and Exporting Finished Products • UK Customers looking for different Suppliers outside the EU • Higher Customs Cost and Longer Shipping Lead time	• Assess the Demand Fluctuation in the UK Compared to the Demand in the EU • Assess if Manufacturing Facilities can be moved to the UK • Assess if the EU Suppliers has the Capacity to Take over the UK Suppliers • Figure out if Direct routes to mainland Europe are alternatives to the UK Land-bridge
Transit through the UK to EU	• Delay in Production and Fulfilment of Customer Orders • Closed border cause miles of queues of trucks (Dover-Calais) route in particular • Higher air freight rates if used as alternative to Maritime routes	• Keeping more Storage of Semi-Finished Products • Paradigm shift from “little and often” supply chain planning • More Inventory against Limited Infrastructure

Brexit Scenario Mapping : Simulation-based Solution

- Mimics the RO/RO Traffic Flow between Ireland, GB, and EU based on the Trade Volumes (Import/Export)
- Model Customs and SPS procedures Post-Brexit



- Maritime Routes (16 Routes)
 - IRE – GB (East/West Corridor)
 - GB – EU-26 (Channel Tunnel)
 - IRE – EU-26 (Land-bridge)
 - IRE – EU-26 (Direct Route)*
- More than 100 Schedules of Vessels Trips

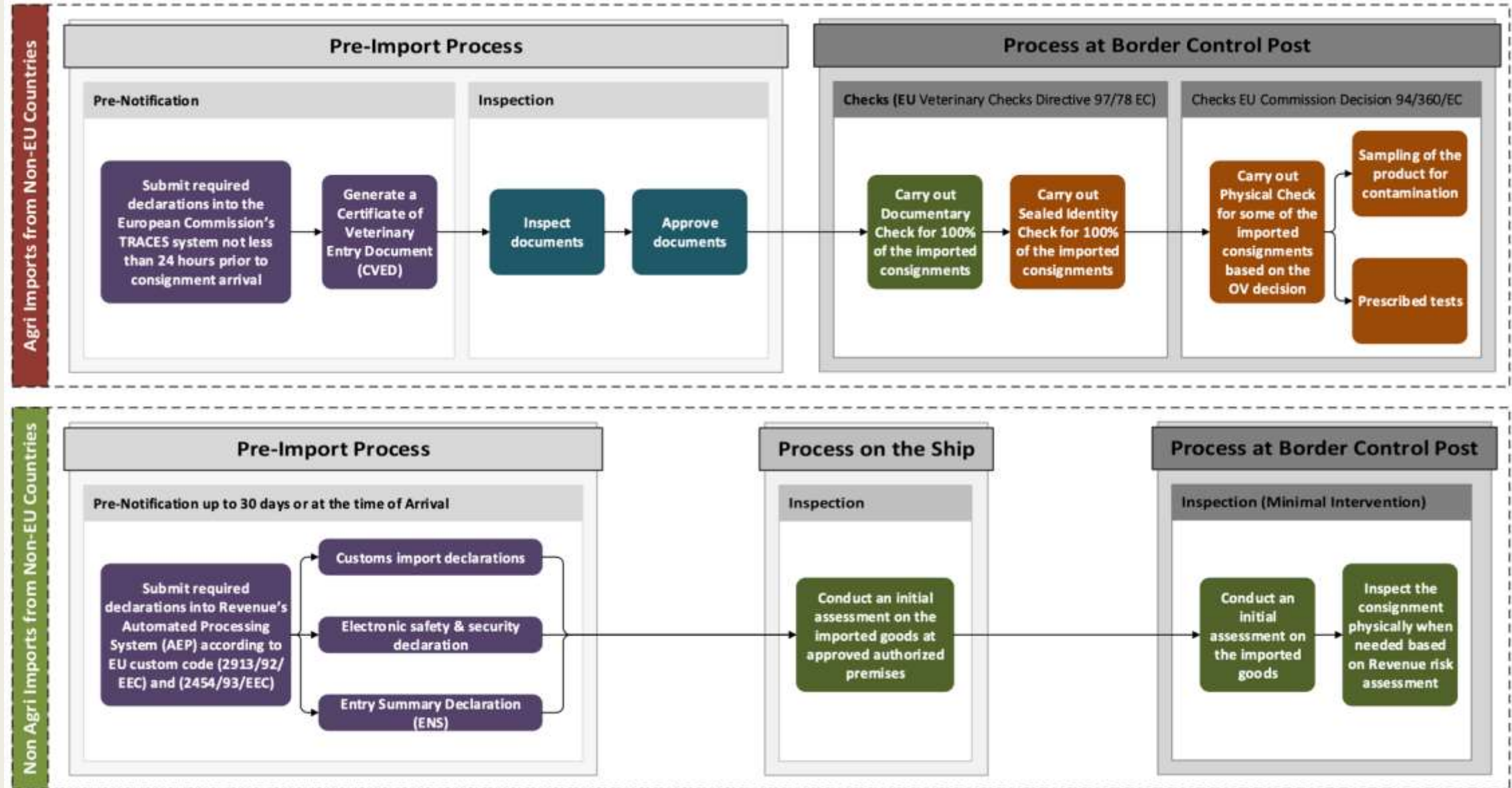
Model Input Parameters

Code	Brexit Scenario Control (BSC)	Parameters
BSC1	Ro/Ro Trade Volume between IRE, GB and EU (All products, Agri-food products, Cheese)	- Ro/Ro trade volume between IRE – UK - Ro/Ro trade volume between IRE – EU (landbridge) - Ro/Ro trade volume between IRE – EU (Direct Route: Cherbourg, Rotterdam, Zeebrugge, and Bilbao) - Ro/Ro trade volume between UK – EU
BSC2	Non-Tariff barriers at IRE, GB and EU Ports	- Checks on outbound/inbound trucks at Irish ports - Checks on outbound/inbound trucks at the UK west ports - Checks on outbound/inbound trucks at the UK east port (Dover) - Checks on outbound/inbound trucks at the EU west ports
BSC3	Checking Resource Capacities at Irish Ports	- Capacities of custom checkpoints at Irish ports (Dublin , Rosslare) - Capacities of SPS checkpoints at Irish ports (Dublin , Rosslare) - Capacities of the immigration and security checkpoints (Dublin, Rosslare)
BSC4	Transportation Modes	Percentage of accompanied/unaccompanied trucks at Irish ports
BSC5	Maritime Routes	Utilisation of Maritime Routes between IRE and the UK
BSC6	Ferries Capacities	- Capacities of Ferries serve IRE – the UK Maritime routes - Capacities of Ferries serve IRE – EU routes

More Than 80 Indicators

- Trucks Transportation Time between IRE, GB, and EU-26 (12 Indicator)
- Products Shelf-life Time (10 Indicator)
- Trucks Waiting Time at Ports' Check Points (20 Indicator)
- Customs & SPS Checkpoints Utilisation (8 Indicator)
- Trucks Queues Length (16 Indicator)
- Border Checking Costs (14 Indicator)

Non-Tariff Barriers



Investigated Scenarios

Non-Tariff Barriers

Limited Checks Intervention

Moderate Checks Intervention

High Checks Intervention

Transit Checks Delay

Limited Transit Delay

Moderate Transit Delay

Long Transit Delay

Demand on Direct Routes

Demand increased 15%

Demand increased 25%

Demand increased 35%

Checkpoints Infrastructure

Lack of Infrastructure at Heysham

Lack of Infrastructure at Holyhead

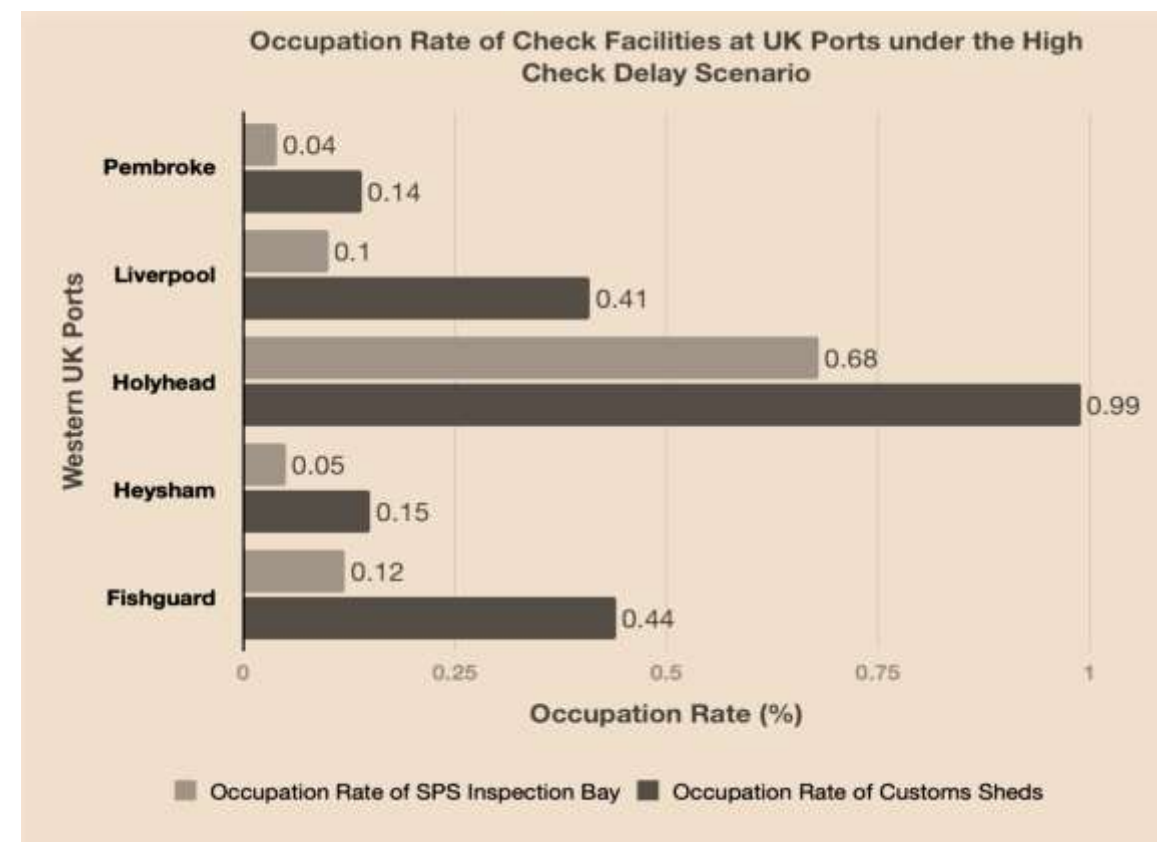
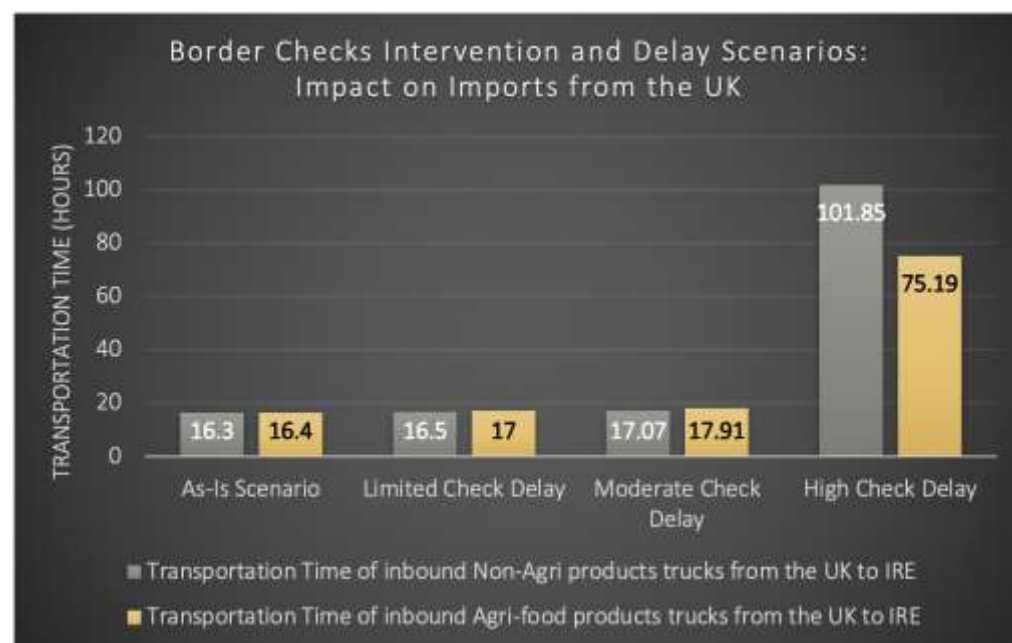
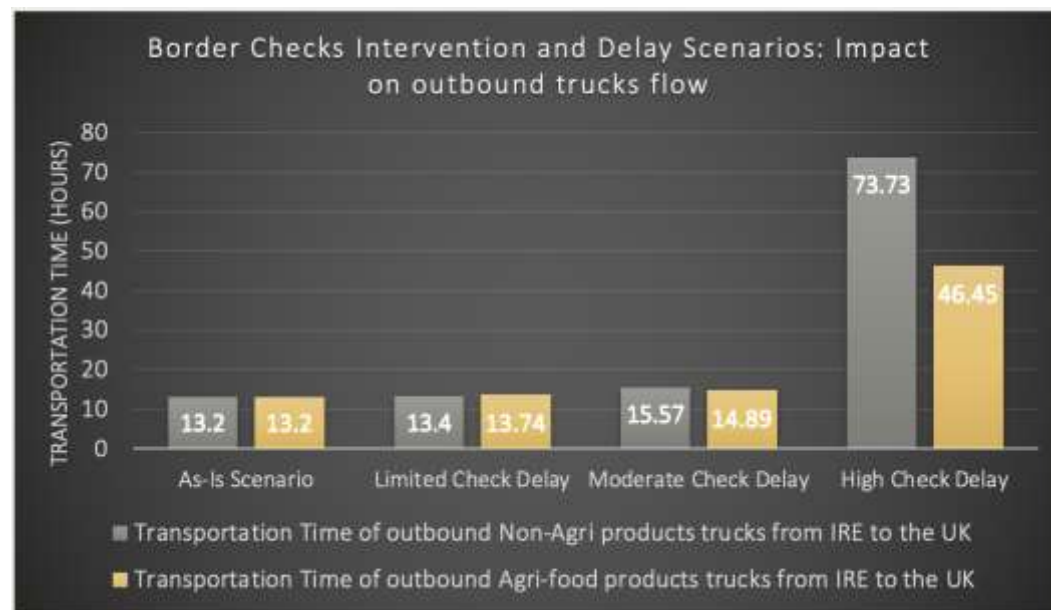
Lack of Infrastructure at Fishguard

Lack of Infrastructure at Pembroke

Non-Tariff Barrier Scenarios

Examined scenarios	Check type	Proportion of trucks (%)	Timing of checks (min)
As-Is Situation	Non-Agri trucks are green-routed (no checks intervention)	100%	0
	Non-Agri trucks are red-routed (physical checks intervention)	0%	0
	Agri-food trucks are physically checked	0%	0
	Trucks selected for immigration and security checks	0%	0
Limited Checks Intervention	Non-Agri trucks are green-routed (no checks intervention)	90%	20
	Non-Agri trucks are red-routed (physical checks intervention)	10%	60
	Agri-food trucks are physically checked (SPS Checks)	10%	60
	Trucks selected for immigration and security checks	10%	20
Moderate Checks Intervention	Non-Agri trucks are green-routed (no checks intervention)	80%	30
	Non-Agri trucks are red-routed (physical checks intervention)	20%	90
	Agri-food trucks are physically checked (SPS Checks)	20%	90
	Trucks selected for immigration and security checks	20%	30
High Checks Intervention	Non-Agri trucks are green-routed (no checks intervention)	70%	45
	Non-Agri trucks are red-routed (physical checks intervention)	30%	135
	Agri-food trucks are physically checked (SPS Checks)	30%	135
	Trucks selected for immigration and security checks	30%	45

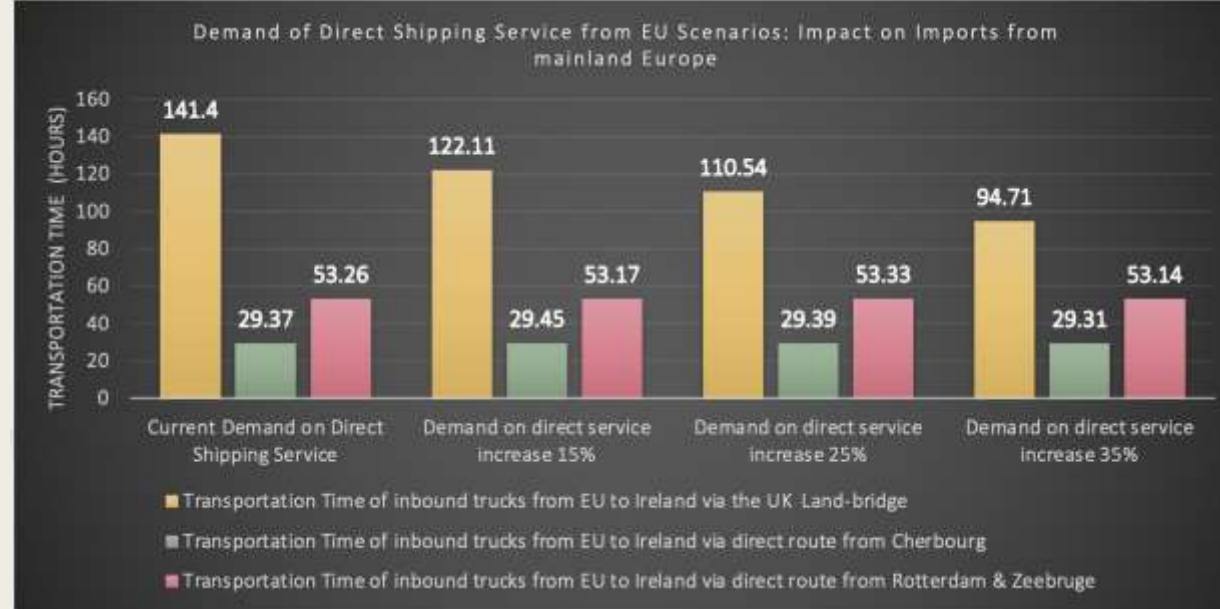
Implications of Non-Tariff Barriers



Direct Route to Europe Vs the UK Land-bridge

	As-Is Scenario	Demand on direct route increase 15%	Demand on direct route increase 25%	Demand on direct route increase 35%
Import Volumes from EU to IRE through Land bridge, Non-Agri (tonnes)	1,205,760	1,024,896	904,320	783,744
Export Volumes from IRE to EU through land bridge, Non-Agri (tonnes)	1,057,920	899,232	793,440	687,648
Import Volumes from EU to IRE through land bridge, Agri-food (tonnes)	555,114	471,846.9	416,336	360,824
Export Volumes from IRE to EU through land bridge, Agri-food (tonnes)	414,629	352,434.65	310,972	269,509
Import Volumes Direct from EU to IRE through Cherbourg, Non-Agri (tonnes)	223,000	283,288	323,480	363,672
Export Volumes Direct from IRE to EU through Cherbourg, Non-Agri (tonnes)	258,000	310,896	346,160	381,424
Import Volumes Direct from EU to IRE through Cherbourg, Agri (tonnes)	173,605	201,361	219,865	238,368
Export Volumes Direct from IRE to EU through Cherbourg, Agri (tonnes)	48,975	69,706	83,527	97,348
Import Volumes Direct from EU to IRE through Rotterdam, Non-Agri (tonnes)	417,000	477,288	517,480	557,672
Export Volumes Direct from EU to IRE through Rotterdam, Non-Agri (tonnes)	256,000	308,896	344,160	379,424
Import Volumes Direct from EU to IRE through Rotterdam, Agri (tonnes)	206,991	234,747	253,251	271,754
Export Volumes Direct from EU to IRE through Rotterdam, Agri (tonnes)	48,975	69,706	83,527	97,348
Import Volumes Direct from EU to IRE through Zeebrugge, Non-Agri (tonnes)	588,000	648,288	688,480	728,672
Export Volumes Direct from EU to IRE through Zeebrugge, Non-Agri (tonnes)	287,000	339,896	375,160	410,424
Import Volumes Direct from EU to IRE through Zeebrugge, Agri (tonnes)	293,793	321,549	340,053	358,556
Export Volumes Direct from EU to IRE through Zeebrugge, Agri (tonnes)	55,097	75,828	89,649	103,470
Import Volumes Direct from EU to IRE through Bilbao, Non-Agri (tonnes)	18,000	78,288	118,480	158,672
Export Volumes Direct from EU to IRE through Bilbao, Non-Agri (tonnes)	0	52,896	88,160	123,424
Import Volumes Direct from EU to IRE through Bilbao, Agri (tonnes)	0	27,756	46,260	64,763
Export Volumes Direct from EU to IRE through Bilbao, Agri (tonnes)	0	20,731	34,552	48,373

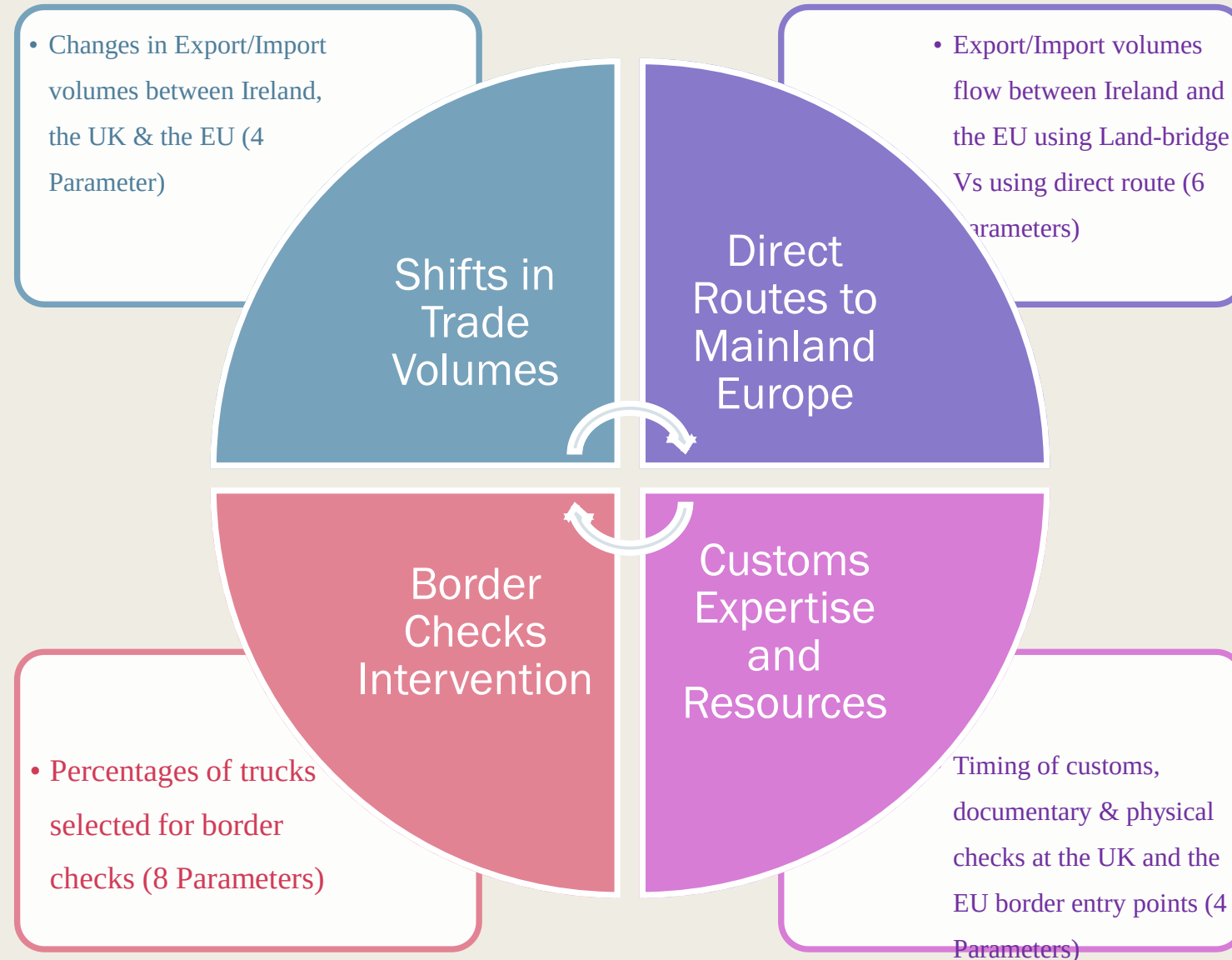
Direct Route to Europe Vs the UK Land-bridge



Findings *(September 2020)*

- In the of High Checks interventions scenario, the transportation time of inbound trucks will increase to up to 3 and 4 days for non-Agri and Agri-food trucks respectively
- Applying SPS Checks on Agri-food trucks could cause increase in the transportation time by 252% for Irish Agri Food Sectors
- A significant bottleneck is anticipated in the link between France and the UK (Calais-Dover). An expected delay from 6 to 9 days is expected to the Inbound flow from mainland Europe to the UK & Ireland
- Dublin-Holyhead is a strategic route links the trade between Ireland and the UK.
- Lack of customs expertise is expected to cause congestions at Ireland, the UK and EU ports.

Post-Brexit Updates (January 2021)



Post-Brexit Deal: Parameters & Indicators

Category	Input Parameter
Shifts in Trade Volume	Import/Export Volumes non-Agri products between GB to Ireland
	Import/Export Volumes Agri products between GB to Ireland
Direct Routes to Mainland Europe	Shift of Import/Export volumes of non-Agri products (from LB to Direct Route route)
	Shift of Import/Export volumes of Agri products (from LB to Direct Route route)
	Land-bridge Vessels Capacity from Dublin and Rosslare to GB
Customs Expertise & Resources	Check timings for documentary and seal identity checks
	Check timings for physical Checks
	Number of Revenue shed for customs checks at Dublin & Rosslare
	Number of SPS check depots at Dublin & Rosslare
Border Checks Intervention	Percentage of Export non-Agri/Agri products trucks selected for pre-bording checks at the Irish ports
	Percentage of Export non-Agri products trucks directed to green/red route at the UK ports
	Percentage of Export Agri product trucks selected for SPS checks at the UK ports
	Percentage of Import non-Agri/Agri products trucks selected for pre-bording checks at the UK ports
	Percentage of Import non-Agri products trucks directed to green route/red route at the Irish ports
	Percentage of Import Agri product trucks selected for SPS checks at the Irish ports



Post-Brexit Deal: Performance Indicators



Category	
Non-Agri Products Flow	<i>Waiting Time of import non-Agri Trucks at the checkpoints in the Irish ports (WT_IB_NA_Dub)/(WT_IB_NA_Ross)</i>
	<i>Waiting time of Export non-Agri Trucks at the checkpoints in the UK Ports (WT_OB_NA_GB-Dub)/(WT_OB_NA_GB-Ross)</i>
Agri Products Flow	<i>Transportation time of import Agri Trucks at the checkpoints in the Irish ports (TT_IB_Agri)</i>
	<i>Transportation time of export Agri Trucks at the checkpoints in the UK ports (TT-OB-Agri)</i>
	<i>Waiting Time of import Agri Trucks at the checkpoints in the Irish ports (WT_IB_A_Dub)/(WT_IB_A_Ross)</i>
	<i>Waiting time of Export Agri Trucks at the checkpoints in the UK Ports (WT_OB_A_GB-Dub)/(WT_OB_A_GB-Ross)</i>
Transportation Routes	<i>Transportation & Waiting time of Import Trucks (Agri and non-Agri) from Ireland to the EU through Land-bridge (TT_IB_LB) / (WT_IB_LB)</i>
	<i>Transportation & Waiting time of Export Trucks (Agri and non-Agri) from the EU to Ireland through Land-bridge (TT_OB_LB)/(WT_OB_LB)</i>
	<i>Transportation time of Import Trucks (Agri and non-Agri) from Ireland to the EU through Direct Route (TT_IB_DR)</i>
	<i>Transportation time of Export Trucks (Agri and non-Agri) from the EU to Ireland through Direct Route (TT_OB_DR)</i>
Checkpoints Utilisation	<i>Occupation rates of Customs checkpoints in the Irish ports (Uti_Cus_D)/(Uti_Cus_R)</i>
	<i>Occupation rates of SPS checkpoints in the Irish ports (Uti_DAEM_D)/(Uti_DAEM_R)</i>

Agri products						Non Agri products				Routes						Staff Utilisation	
OutBound		Inbound				Dublin		Rooslare		Land bridge				Direct Route		Dublin	
TT	WT	WT	WT	WT	WT	WT (Inbound)	WT (Outbound GB)	WT (Inbound)	WT (Outbound GB)	TT (Outbound)	WT (Outbound)	TT (Inbound)	WT (Inbound)	TT (Outbound)	TT (Inbound)	Customs	DAFM
0.66	0.00	0.97	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.16	0.00	0.00	0.00	0.01	0.00	0.00	0.00
0.99	0.96	0.00	0.87	0.97	0.00	0.00	0.77	0.07	0.95	0.39	0.00	0.00	0.00	0.60	0.00	0.00	0.00
0.67	0.84	0.92	0.59	0.42	0.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.45	0.00	0.73	0.00	0.00
0.00	0.97	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.92	0.00	0.83	0.00	0.00
0.00	0.00	0.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.91	0.00
0.66	0.00	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.64	0.00	0.00	0.00
0.73	0.00	0.82	0.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.81	0.93	0.00	0.00
0.00	0.84	0.79	0.77	0.69	0.67	0.00	0.00	0.00	0.00	0.71	0.83	0.00	0.00	0.99	0.00	0.00	0.96
0.85	0.00	0.00	0.89	0.00	0.00	0.00	0.00	0.00	0.00	0.52	0.09	0.00	0.00	0.69	0.72	0.00	0.00
0.00	0.00	0.00	0.45	0.72	0.80	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.00	0.75	0.07	0.00	0.00
0.00	0.99	0.78	0.69	0.83	0.88	0.00	0.87	0.05	0.38	0.00	0.72	0.00	0.03	0.80	0.07	0.00	0.79
0.00	0.74	0.95	0.98	0.00	0.00	0.73	0.45	0.59	0.00	0.15	0.70	0.66	0.76	0.50	0.52	0.00	0.00
0.80	0.82	0.86	0.77	0.86	0.82	0.60	0.00	0.00	0.00	0.12	0.00	0.62	0.47	0.08	0.52	0.00	0.77
0.99	0.00	0.63	0.96	0.82	0.92	0.75	0.00	0.00	0.00	0.51	0.70	0.68	0.53	0.07	0.00	0.00	0.76
0.87	0.00	0.00	0.67	0.81	0.82	0.06	0.61	0.36	0.84	0.17	0.80	0.00	0.00	0.12	0.00	0.00	0.92
0.87	0.00	0.00	0.99	0.96	0.79	0.48	0.03	0.62	0.00	0.96	0.68	0.47	0.47	0.57	0.00	0.00	0.98
0.00	0.74	0.54	0.78	0.92	0.00	0.72	0.87	0.39	0.51	0.58	0.82	0.94	0.90	0.62	0.65	0.00	0.00
0.00	0.00	0.57	0.00	0.00	0.87	0.06	0.50	0.49	0.38	0.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.66	0.82	0.64	0.58	0.71	0.00	0.96	0.71	0.82	0.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PREDICTORS/DETERMINANTS OF THE KEY PERFORMANCE MEASURES																	
2	5	3	2	5	5	7	6	4	5	2	5	6	7	2	2	6	6

Data collection: DoE

Generating nearly orthogonal Latin Hypercube designs

- NOLH designs: (nearly) orthogonal designs (see Cioppa and Lucas 2007).
- Good space-filling properties
- Nearly orthogonal; all correlations between the columns in the design matrix are in the interval $(-.03, .03)$
- The algorithm takes extensive time to generate these designs;
- A catalogue of ready-to-use, NOLH with good space-filling designs for up to 22 factors in as few as 129 runs
- 19 continuous parameters, represented as percentage of change (lower and upper bounded)
- Experimental Design tool can be accessed from this link
<https://www.nps.edu/web/seed/software-downloads>

Predictors/determinants of the key performance measures

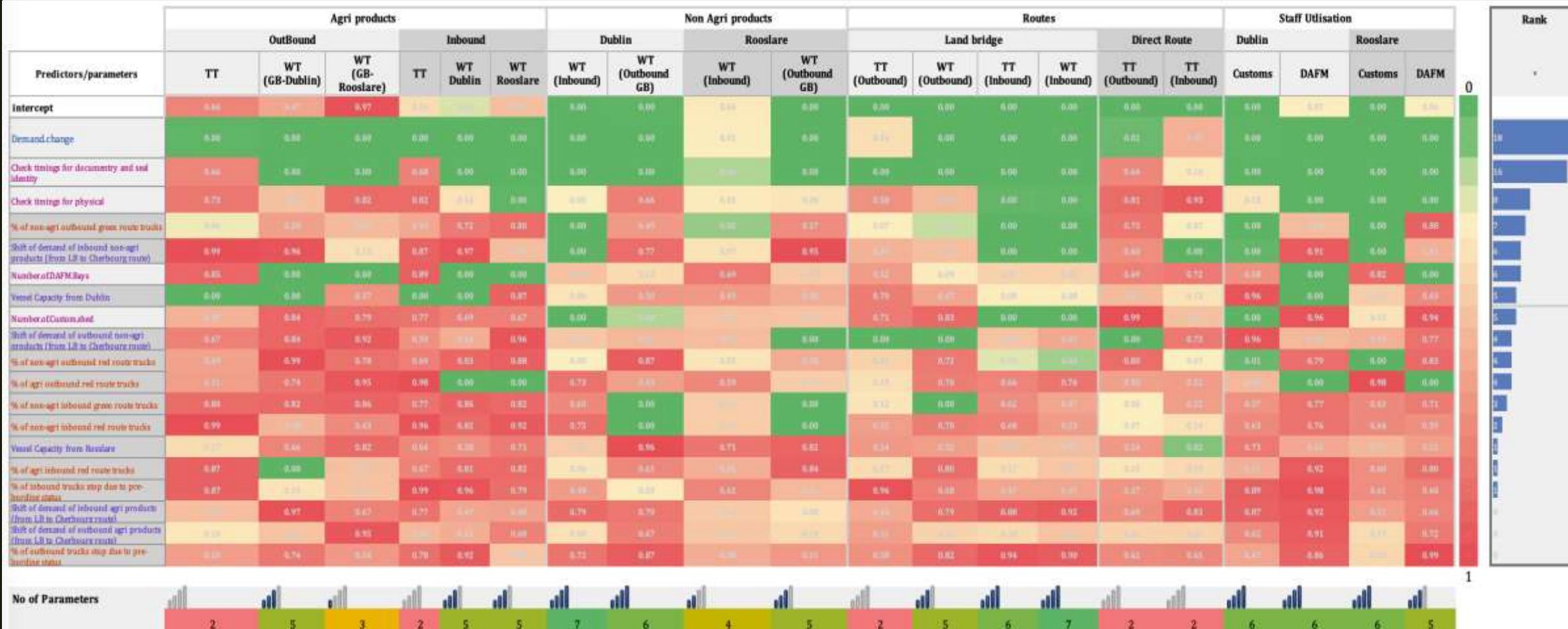
LM Models (param- P-value) visualization



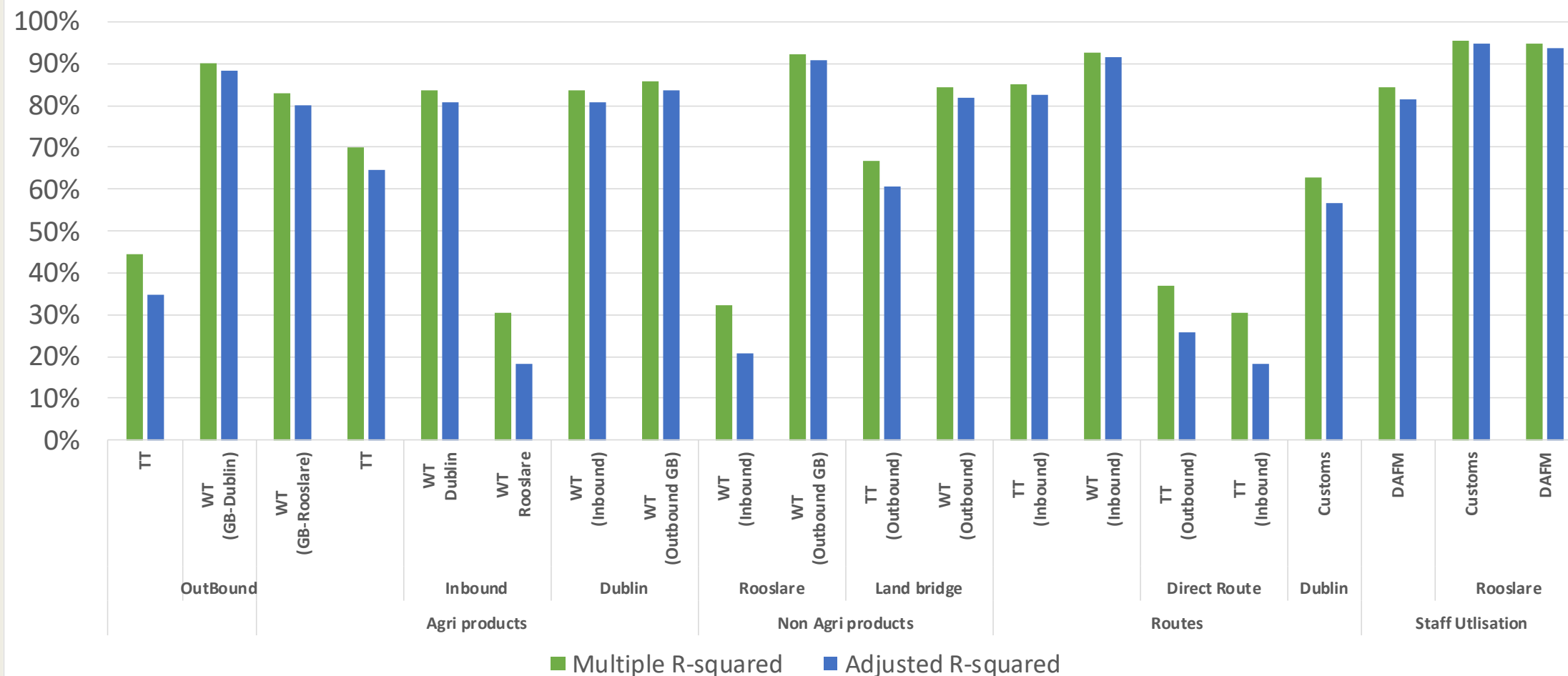
	Predictors/parameters	Agri products						Non Agri products				Routes						Staff Utilisation				0
		OutBound			Inbound			Dublin		Rooslare		Land bridge				Direct Route		Dublin		Rooslare		
		TT	WT (GB-Dublin)	WT (GB-Rooslare)	TT	WT Dublin	WT Rooslare	WT (Inbound)	WT (Outbound GB)	WT (Inbound)	WT (Outbound GB)	TT (Outbound)	WT (Outbound)	TT (Inbound)	WT (Inbound)	TT (Outbound)	TT (Inbound)	Customs	DAFM	Customs	DAFM	
Direct Routes to mainland europe	Intercept	0.66	0.77	0.97	0.76	0.99	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.00	
	Shift of demand of inbound non-agri products (from LB to Cherbourg route)	0.99	0.96	0.75	0.87	0.97	0.99	0.00	0.77	0.07	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.91	0.00	0.00	
	Shift of demand of outbound non-agri products (from LB to Cherbourg route)	0.87	0.84	0.92	0.33	0.80	0.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.71	0.96	0.00	0.00	0.77		
	Shift of demand of inbound agri products (from LB to Cherbourg route)	0.00	0.97	0.62	0.77	0.67	0.84	0.79	0.70	0.00	0.00	0.54	0.79	0.88	0.92	0.00	0.61	0.87	0.92	0.55	0.66	
	Shift of demand of outbound agri products (from LB to Cherbourg route)	0.00	0.00	0.95	0.00	0.00	0.68	0.00	0.07	0.00	0.54	0.21	0.22	0.00	0.00	0.22	0.22	0.62	0.91	0.17	0.72	
	Vessel Capacity from Dublin	0.00	0.00	0.07	0.00	0.00	0.87	0.00	0.00	0.00	0.00	0.70	0.07	0.00	0.00	0.00	0.21	0.96	0.00	0.00	0.63	
	Vessel Capacity from Rooslare	0.77	0.66	0.82	0.64	0.00	0.71	0.00	0.96	0.71	0.82	0.34	0.22	0.00	0.00	0.20	0.02	0.73	0.00	0.00	0.02	
Customs Inspections and Resources	Check timings for documentary and seal identity	0.66	0.00	0.00	0.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	Check timings for physical	0.73	0.00	0.02	0.82	0.14	0.00	0.00	0.66	0.03	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	Number of Customs staff	0.73	0.84	0.79	0.77	0.69	0.67	0.00	0.00	0.00	0.00	0.71	0.83	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00	
	Number of DAFM staff	0.85	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.52	0.00	0.00	0.00	0.00	0.71	0.00	0.00	0.00	0.00	
Border checks interventions	% of non-agri outbound green route trucks	0.00	0.00	0.00	0.13	0.72	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.71	0.00	0.00	0.00	0.00	0.00	
	% of non-agri outbound red route trucks	0.00	0.00	0.70	0.60	0.03	0.00	0.00	0.07	0.00	0.00	0.21	0.71	0.00	0.00	0.00	0.00	0.79	0.00	0.00	0.03	
	% of agri outbound red route trucks	0.00	0.74	0.95	0.98	0.00	0.00	0.73	0.00	0.00	0.00	0.00	0.70	0.00	0.75	0.52	0.52	0.00	0.00	0.98	0.00	
	% of non-agri inbound green route trucks	0.00	0.02	0.00	0.77	0.00	0.02	0.00	0.00	0.00	0.13	0.00	0.00	0.00	0.00	0.00	0.22	0.57	0.77	0.00	0.71	
	% of non-agri inbound red route trucks	0.00	0.00	0.00	0.96	0.02	0.92	0.73	0.00	0.00	0.21	0.70	0.00	0.00	0.00	0.00	0.21	0.00	0.70	0.00	0.00	
	% of agri inbound red route trucks	0.07	0.00	0.00	0.07	0.01	0.02	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.12	0.00	0.00	0.00	0.00	0.00	
	% of inbound trucks stop due to pre-clearing status	0.07	0.11	0.00	0.96	0.00	0.79	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	% of outbound trucks stop due to pre-clearing status	0.00	0.00	0.00	0.70	0.02	0.00	0.72	0.07	0.00	0.00	0.20	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
Shift in Trade Volume	Demand change	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	

Predictors/determinants of the key performance measures

LM Models (param- P-value) visualization + Ranking

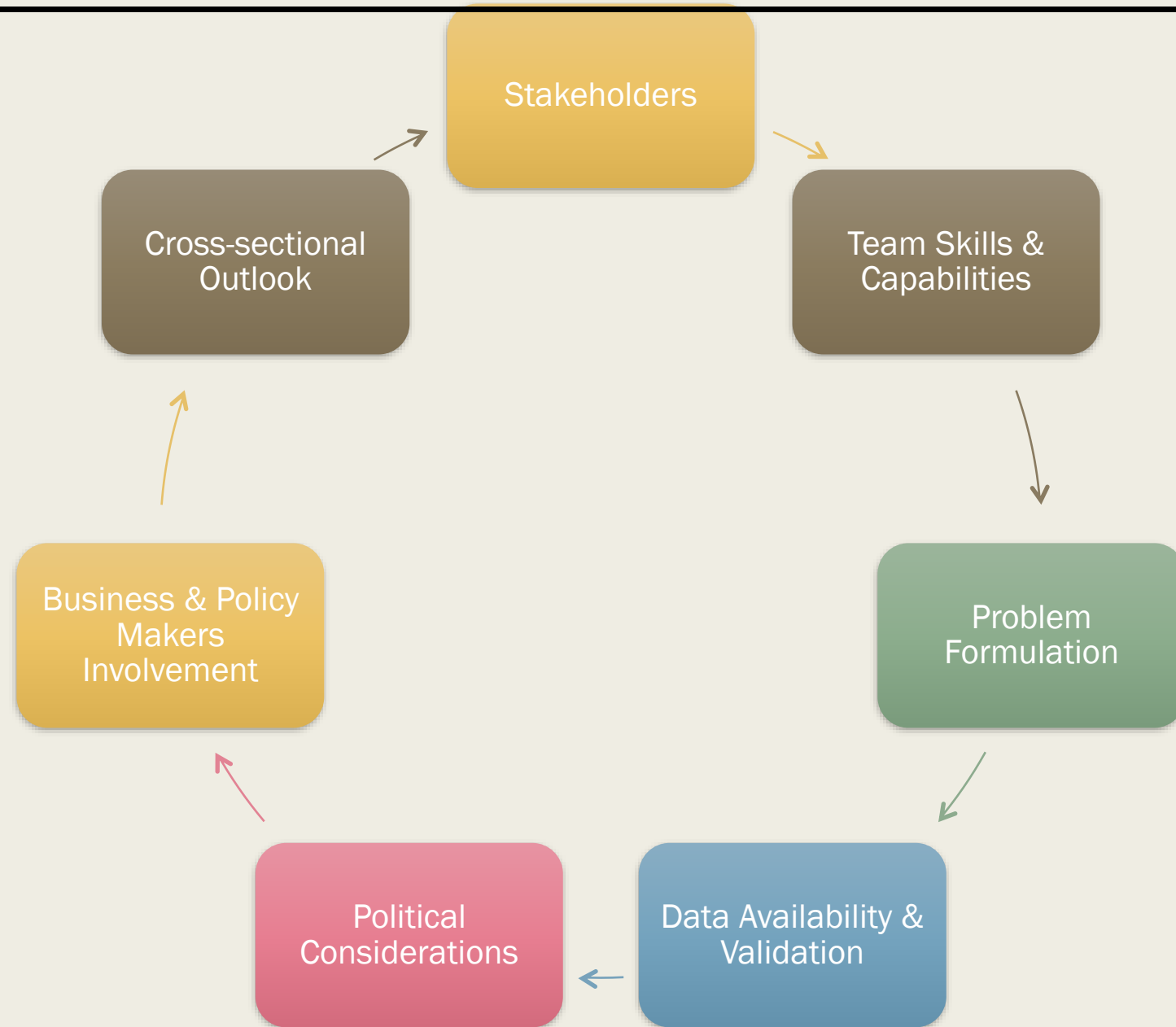


% of variations explained by the LM models for each KPI



- Realise the expected collapse in Demand
- Limit level of checks intervention at the ports.
- Hauliers to be familiar with all customs digital solution.
- Adequate inspection facilities are essential.
- Border Inspection guidance must be published.
- Apply for authorised consignee status (i.e. Authorised Economic Operator).
- Come up with best operational and economic models of direct shipping services to mainland Europe.
- Prepare for a surge in demand for refrigerated vans and trucks (reefers).
- Addition of shipping capacities and use of Rosslare port.

Simulation Role in Business Resilience: Challenges & Enablers



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Thank you
Enjoy the Conference!